



# Contraindications and Precautions

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For Health Care Providers

## KEY POINTS

- Contraindications to vaccination are conditions that increase the risk for a serious adverse reaction. Vaccines should not be administered when a contraindication is present.
- Precautions are conditions that might increase the risk for a serious adverse reaction, cause diagnostic confusion, or compromise the ability of the vaccine to produce immunity.
- In general, vaccinations should be deferred when a precaution is present. However, vaccination might still be indicated if the benefit outweighs the risk.

## Updates

Major changes to the best practice guidance in this section include 1) enhancement of the definition of a “precaution” to include any condition that might confuse diagnostic accuracy and 2) recommendation to vaccinate during a hospitalization if a patient is not acutely moderately or severely ill.

## General Principles

National standards for pediatric vaccination practices have been established and include descriptions of valid contraindications and precautions to vaccination [\[1\]](#). Persons who administer vaccines should screen patients for contraindications and precautions to the vaccine before each dose of vaccine is administered (Table 4-1). Screening is facilitated by consistent use of screening questionnaires, which are available from certain state vaccination programs and other sources (e.g., the [Immunization Action Coalition](#) [\[2\]](#)).

## Contraindications

Contraindications (conditions in a recipient that increases the risk for a serious adverse reaction) to vaccination are conditions under which vaccines should not be administered. Because the majority of contraindications are temporary, vaccinations often can be administered later when the condition leading to a contraindication no longer exists. A vaccine should not be administered when a contraindication is present; for example, MMR vaccine should not be administered to severely immunocompromised persons [\[2\]](#). However, certain conditions are commonly misperceived as contraindications (i.e., are not valid reasons to defer vaccination).

Severely immunocompromised persons generally should not receive live vaccines [\[3\]](#). Because of the theoretical risk to the fetus, women known to be pregnant generally should not receive live, attenuated virus vaccines [\[4\]](#). Persons who experienced encephalopathy within 7 days after administration of a previous dose of pertussis-containing vaccine not attributable to another identifiable cause should not receive additional doses of a vaccine that contains pertussis [\[4\]](#) [\[5\]](#). Severe Combined Immunodeficiency (SCID) disease and a history of intussusception are both contraindications to the receipt of rotavirus vaccines [\[6\]](#).

## Precautions

A precaution is a condition in a recipient that might increase the risk for a serious adverse reaction, might cause diagnostic confusion, or might compromise the ability of the vaccine to produce immunity (e.g., administering measles vaccine to a person with passive immunity to measles from a blood transfusion administered up to 7 months prior) [\[7\]](#). A person might experience a more severe reaction to the vaccine than would have otherwise been expected; however, the risk for this happening is less than the risk expected with a contraindication. In general, vaccinations should be deferred when a precaution is present. However, a vaccination might be indicated in the presence of a precaution if the benefit of protection from the vaccine outweighs the risk for an adverse reaction.

The presence of a moderate or severe acute illness with or without a fever is a precaution to administration of all vaccines (Table 4-1). The decision to administer or delay vaccination because of a current or recent acute illness depends on the severity of symptoms and etiology of the condition. The safety and efficacy of vaccinating persons who have mild illnesses have been documented [8] [9] [10] [11]. Vaccination should be deferred for persons with a moderate or severe acute illness. This precaution avoids causing diagnostic confusion between manifestations of the underlying illness and possible adverse effects of vaccination or superimposing adverse effects of the vaccine on the underlying illness. After they are screened for contraindications, persons with moderate or severe acute illness should be vaccinated as soon as the acute illness has improved. Studies indicate that failure to vaccinate children with minor illnesses can impede vaccination efforts [12] [13] [14]. Among persons whose compliance with medical care cannot be ensured, use of every opportunity to administer appropriate vaccines is critical.

Hospitalization should be used as an opportunity to provide recommended vaccinations. Health-care facilities are held to standards of offering influenza vaccine for hospitalized patients, so providers are incentivized to vaccinate these patients at some point during hospitalization [15]. Likewise, patients admitted for elective procedures will not be acutely ill during all times during their hospitalization. Most studies that have explored the effect of surgery or anesthesia on the immune system were observational, included only infants and children, and were small and indirect, in that they did not look at the immune effect on the response to vaccination specifically [16] [17] [18] [19] [20] [21] [22] [23] [24] [25] [26] [27] [28] [29] [30] [31] [32] [33] [34] [35]. They do not provide convincing evidence that recent anesthesia or surgery significantly affect response to vaccines. Current, recent, or upcoming anesthesia/surgery/hospitalization is not a contraindication to vaccination, but certain factors might lead a provider to consider current, recent, or upcoming anesthesia/surgery/hospitalization as a precaution [16] [17] [18] [19] [20] [21] [22] [23] [24] [25] [26] [27] [28] [29] [30] [31] [32] [33] [34] [35]. Efforts should be made to ensure vaccine administration during the hospitalization or at discharge. For patients who are deemed moderately or severely ill throughout the hospitalization, vaccination should occur at the earliest opportunity (i.e., during immediate post-hospitalization follow-up care, including home or office visits) when patients' clinical symptoms have improved.

A personal or family history of seizures is a precaution for MMRV vaccination; this is because a recent study found an increased risk for febrile seizures in children 12-23 months who receive MMRV compared with MMR and varicella vaccine [36].

## Neither Contraindications nor Precautions

Clinicians or other health-care providers might misperceive certain conditions or circumstances as valid contraindications or precautions to vaccination when they actually do not preclude vaccination [1] (Table 4-2). These misperceptions result in missed opportunities to administer recommended vaccines [37].

Routine physical examinations and procedures (e.g., measuring temperatures) are not prerequisites for vaccinating persons who appear to be healthy. The provider should ask the parent or guardian if the child is ill. If the child has a moderate or severe illness, the vaccination should be postponed.

TABLE 4-1. Contraindications and precautions<sup>(a)</sup> to commonly used vaccines

References in this table: [2] [4] [6] [38] [39] [40] [41] [42] [43] [44] [45] [46] [47] [48] [49] [50] [51] [52] [53]

| Vaccine                                                                                                                                     | Citation | Contraindications                                                                                                                                                                                                                                                                                                                                                                                            | Precautions                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dengue— ONLY use in persons who have laboratory confirmation of previous dengue infection AND reside in endemic dengue areas <sup>(b)</sup> | [38]     | Lack of laboratory evidence of previous dengue infection<br><br>Severe allergic reaction (e.g., anaphylaxis) after a previous dose or to a vaccine component<br><br>Severe immunodeficiency (e.g., hematologic and solid tumors, receipt of chemotherapy, congenital immunodeficiency, long-term immunosuppressive therapy <sup>(c)</sup> or patients with HIV infection who are severely immunocompromised) | Pregnancy<br><br>HIV infection without evidence of severe immunosuppression<br><br>Moderate or severe acute illness with or without fever                                                                                                                                                                                                                                                                                             |
| DT, Td                                                                                                                                      | [4]      | Severe allergic reaction (e.g., anaphylaxis) after a previous dose or to a vaccine component                                                                                                                                                                                                                                                                                                                 | GBS <6 weeks after previous dose of tetanus-toxoid-containing vaccine<br><br>History of Arthus-type hypersensitivity reactions after a previous dose of diphtheria-toxoid—containing or tetanus-toxoid-containing vaccine; defer vaccination until at least 10 years have elapsed since the last tetanus-toxoid-containing vaccine<br><br>Moderate or severe acute illness with or without fever                                      |
| DTaP                                                                                                                                        | [39]     | Severe allergic reaction (e.g., anaphylaxis) after a previous dose or to a vaccine component<br>Encephalopathy (e.g., coma, decreased level of consciousness, prolonged seizures), not attributable to another identifiable cause, within 7 days of administration of previous dose of DTP or DTaP                                                                                                           | Progressive neurologic disorder, including infantile spasms, uncontrolled epilepsy, progressive encephalopathy; defer DTaP until neurologic status clarified and stabilized<br><br>GBS <6 weeks after previous dose of tetanus-toxoid-containing vaccine<br><br>History of Arthus-type hypersensitivity reactions after a previous dose of diphtheria-toxoid-containing or tetanus-toxoid-containing vaccine; defer vaccination until |

| Vaccine                | Citation                 | Contraindications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Precautions                                                                                                                                                                                                                                                                                                 |
|------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        |                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>at least 10 years have elapsed since the last tetanus-toxoid-containing vaccine</p> <p>Moderate or severe acute illness with or without fever</p>                                                                                                                                                        |
| Hepatitis A            | <a href="#">(40)</a>     | Severe allergic reaction (e.g., anaphylaxis) after a previous dose or to a vaccine component                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Moderate or severe acute illness with or without fever                                                                                                                                                                                                                                                      |
| Hepatitis B            | <a href="#">(41)</a>     | Severe allergic reaction (e.g., anaphylaxis) after a previous dose or to a vaccine component<br>Hypersensitivity to yeast                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Moderate or severe acute illness with or without fever                                                                                                                                                                                                                                                      |
| Hib                    | <a href="#">(42)</a>     | Severe allergic reaction (e.g., anaphylaxis) after a previous dose or to a vaccine component<br>Age <6 weeks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Moderate or severe acute illness with or without fever                                                                                                                                                                                                                                                      |
| HPV <sup>(d)</sup>     | <a href="#">(43)</a>     | Severe allergic reaction (e.g., anaphylaxis) after a previous dose or to a vaccine component, including yeast                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Moderate or severe acute illness with or without fever                                                                                                                                                                                                                                                      |
| IIV <sup>(e)</sup>     | <a href="#">(44)</a>     | Severe allergic reaction (e.g., anaphylaxis) after previous dose of influenza vaccine or to vaccine component                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>GBS &lt;6 weeks after a previous dose of influenza vaccine</p> <p>Moderate or severe acute illness with or without fever</p>                                                                                                                                                                             |
| IPV                    | <a href="#">(45)</a>     | Severe allergic reaction (e.g., anaphylaxis) after a previous dose or to a vaccine component                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>Pregnancy</p> <p>Moderate or severe acute illness with or without fever</p>                                                                                                                                                                                                                              |
| LAIV <sup>(f)</sup>    | <a href="#">(44)</a>     | <p>Severe allergic reaction (e.g., anaphylaxis) after a previous dose or to a vaccine component</p> <p>Concomitant use of aspirin or salicylate- containing medication in children and adolescents</p> <p>LAIV4 should not be administered to persons who have taken oseltamivir or zanamivir within the previous 48 hours, peramivir within the previous 5 days, or baloxavir within the previous 17 days.<sup>(h)</sup></p> <p>Pregnancy</p> <p>Children aged 2 through 4 years who have received a diagnosis of asthma or whose parents or caregivers report that a health care provider has told them during the preceding 12 months that their child had wheezing or asthma or whose medical record indicates a wheezing episode has occurred during the preceding 12 months.</p> <p>Persons with active cerebrospinal fluid/oropharyngeal communications/leaks.</p> <p>Close contacts and caregivers of severely immunosuppressed persons who require a protected environment.</p> <p>Persons with cochlear implants (due to the potential for CSF leak, which might exist for some period of time after implantation. Providers might consider consultation with a specialist concerning risk of persistent CSF leak if an age-appropriate inactivated or recombinant vaccine cannot be used).</p> <p>Altered Immunocompetence</p> <p>Anatomic or functional asplenia (e.g. sickle cell disease)</p> | <p>GBS &lt;6 weeks after a previous dose of influenza vaccine</p> <p>Asthma in persons aged 5 years old or older</p> <p>Medical conditions which might predispose to higher risk of complications attributable to influenza<sup>(g)</sup></p> <p>Moderate or severe acute illness with or without fever</p> |
| MenACWY                | <a href="#">(46)</a>     | Severe allergic reaction (e.g., anaphylaxis) after a previous dose or to a vaccine component                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>Moderate or severe acute illness with or without fever</p> <p>Preterm birth (MenACWY-CRM)<sup>(i)</sup></p>                                                                                                                                                                                              |
| MenB                   | <a href="#">(46, 45)</a> | Severe allergic reaction (e.g., anaphylaxis) after a previous dose or to a vaccine component                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>Moderate or severe acute illness with or without fever</p> <p>Pregnancy</p> <p>Latex sensitivity (MenB-4c)</p>                                                                                                                                                                                           |
| MMR <sup>(j),(k)</sup> | <a href="#">(1)</a>      | <p>Severe allergic reaction (e.g., anaphylaxis) after a previous dose or to a vaccine component</p> <p>Pregnancy</p> <p>Known severe immunodeficiency (e.g., from hematologic and solid tumors, receipt of chemotherapy, congenital immunodeficiency, long-term</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>Recent (≤11 months) receipt of antibody-containing blood product (specific interval depends on product)</p> <p>History of thrombocytopenia or thrombocytopenic purpura</p>                                                                                                                               |

| Vaccine                      | Citation             | Contraindications                                                                                                                                                                                                                                                                                                                                                                                                                                            | Precautions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              |                      | <p>immunosuppressive therapy<sup>(c)</sup> or patients with HIV infection who are severely immunocompromised<sup>(j)</sup></p> <p>Family history of altered immunocompetence<sup>(m)</sup></p>                                                                                                                                                                                                                                                               | <p>Need for tuberculin skin testing or interferon-gamma release assay (IGRA) testing<sup>(l)</sup></p> <p>Moderate or severe acute illness with or without fever</p>                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| MPSV4                        | <a href="#">(49)</a> | Severe allergic reaction (e.g., anaphylaxis) after a previous dose or to a vaccine component                                                                                                                                                                                                                                                                                                                                                                 | Moderate or severe acute illness with or without fever                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| PCV13, PCV15, PCV20          | <a href="#">(50)</a> | Severe allergic reaction (e.g., anaphylaxis) after a previous dose of PCV or any diphtheria-toxoid–containing vaccine or to a component of a vaccine (PCV or any diphtheria-toxoid–containing vaccine)                                                                                                                                                                                                                                                       | Moderate or severe acute illness with or without fever                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| PPSV23                       | <a href="#">(51)</a> | Severe allergic reaction (e.g., anaphylaxis) after a previous dose or to a vaccine component                                                                                                                                                                                                                                                                                                                                                                 | Moderate or severe acute illness with or without fever                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| RIV                          | <a href="#">(44)</a> | Severe allergic reaction (e.g., anaphylaxis) to any component of the vaccine                                                                                                                                                                                                                                                                                                                                                                                 | <p>GBS &lt;6 weeks after a previous dose of influenza vaccine</p> <p>Moderate or severe acute illness with or without fever</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Rotavirus                    | <a href="#">(6)</a>  | <p>Severe allergic reaction (e.g., anaphylaxis) after a previous dose or to a vaccine component</p> <p>SCID</p> <p>History of intussusception</p>                                                                                                                                                                                                                                                                                                            | <p>Altered immunocompetence other than SCID</p> <p>Chronic gastrointestinal disease</p> <p>Spina bifida or bladder exstrophy<sup>(n)</sup></p> <p>Moderate or severe acute illness with or without fever</p>                                                                                                                                                                                                                                                                                                                                                                                            |
| Tdap                         | <a href="#">(52)</a> | <p>Severe allergic reaction (e.g., anaphylaxis) after a previous dose or to a vaccine component</p> <p>Encephalopathy (e.g., coma, decreased level of consciousness, prolonged seizures), not attributable to another identifiable cause, within 7 days of administration of previous dose of DTP, DTaP, or Tdap</p>                                                                                                                                         | <p>GBS &lt;6 weeks after a previous dose of tetanus-toxoid–containing vaccine</p> <p>Progressive or unstable neurological disorder, uncontrolled seizures, or progressive encephalopathy until a treatment regimen has been established and the condition has stabilized</p> <p>History of Arthus-type hypersensitivity reactions after a previous dose of diphtheria-toxoid—containing or tetanus-toxoid–containing vaccine; defer vaccination until at least 10 years have elapsed since the last tetanus-toxoid–containing vaccine</p> <p>Moderate or severe acute illness with or without fever</p> |
| Varicella <sup>(i),(k)</sup> | <a href="#">(53)</a> | <p>Severe allergic reaction (e.g., anaphylaxis) after a previous dose or to a vaccine component</p> <p>Known severe immunodeficiency (e.g., from hematologic and solid tumors, receipt of chemotherapy, congenital immunodeficiency, long-term immunosuppressive therapy<sup>(c)</sup> or patients with HIV infection who are severely immunocompromised<sup>(j)</sup>)</p> <p>Pregnancy</p> <p>Family history of altered immunocompetence<sup>(m)</sup></p> | <p>Recent (≤11 months) receipt of antibody-containing blood product (specific interval depends on product)</p> <p>Moderate or severe acute illness with or without fever</p> <p>Receipt of specific antiviral drugs (acyclovir, famciclovir, or valacyclovir) 24 hours before vaccination (avoid use of these antiviral drugs for 14 days after vaccination)</p> <p>Use of aspirin or aspirin-containing products<sup>(o)</sup></p>                                                                                                                                                                     |
| Zoster                       | <a href="#">(54)</a> | Severe allergic reaction (e.g., anaphylaxis) after a previous dose or to a vaccine component                                                                                                                                                                                                                                                                                                                                                                 | Moderate or severe acute illness with or without fever                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

**Abbreviations:** DT = diphtheria and tetanus toxoids; DTaP = diphtheria and tetanus toxoids and acellular pertussis; DTP = diphtheria toxoid, tetanus toxoid, and pertussis; GBS = Guillain-Barré syndrome; Hib = Haemophilus influenzae type b; HIV = human immunodeficiency virus; HPV = human papillomavirus; IIV = inactivated influenza vaccine; IPV = inactivated poliovirus; LAIV = live, attenuated influenza vaccine; MenACWY = quadrivalent meningococcal conjugate vaccine; MMR = measles, mumps, and rubella; MPSV4 = quadrivalent meningococcal polysaccharide vaccine; PCV13 = pneumococcal conjugate vaccine; PPSV23= pneumococcal polysaccharide vaccine; SCID = severe combined immunodeficiency; RIV=recombinant influenza vaccine; Td = tetanus and diphtheria toxoids; Tdap = tetanus toxoid, reduced diphtheria toxoid, and acellular pertussis.

<sup>(a)</sup> Events or conditions listed as precautions should be reviewed carefully. Benefits of and risks for administering a specific vaccine to a person under these circumstances should be considered. If the risk from the vaccine is believed to outweigh the benefit, the vaccine should not be administered. If the benefit of vaccination is believed to outweigh the risk, the vaccine should be administered. Whether and when to administer DTaP to children with proven or suspected underlying neurologic disorders should be decided on a case-by-case basis.

<sup>(b)</sup> Only persons with laboratory confirmation of immunity according to strict guidance at [Laboratory Testing Requirements for Vaccination with Dengvaxia Dengue Vaccine](#) should receive dengue vaccination.

<sup>(c)</sup> Substantially immunosuppressive steroid dose is considered to be  $\geq 2$  weeks of daily receipt of 20 mg or 2 mg/kg body weight of prednisone or equivalent.

<sup>(d)</sup> HPV vaccine is not recommended during pregnancy

<sup>(e)</sup> When applying this contraindication to cclIV, the history of severe allergic reaction (e.g., anaphylaxis) must be specific to the event occurring following a dose of cclIV. Likewise, when applying this contraindication to RIV, the history of severe allergic reaction (e.g., anaphylaxis) must be specific to the event occurring following a dose of RIV. A history of severe allergic reaction (e.g., anaphylaxis) to a non-cclIV vaccine or to a component specific to components not contained in cclIV, is a precaution to cclIV. A history of severe allergic reaction (e.g., anaphylaxis) to a non-RIV vaccine or to a component specific to components not contained in RIV is a precaution to RIV.

<sup>(f)</sup> In addition, ACIP recommends LAIV not be used for pregnant women, immunosuppressed persons, and children aged 2-4 years who have asthma or who have had a wheezing episode noted in the medical record within the past 12 months, or for whom parents report that a health care provider stated that they had wheezing or asthma within the last 12 months. LAIV should not be administered to persons who have taken influenza antiviral medications within the previous 48 hours. Persons who care for severely immunosuppressed persons who require a protective environment should not receive LAIV, or should avoid contact with such persons for 7 days after receipt.

<sup>(g)</sup> See reference: Grohskopf LA, Alyanak E, Ferdinands JM, et al. Prevention and Control of Seasonal Influenza with Vaccines: Recommendations of the Advisory Committee on Immunization Practices, United States, 2021-2022 Influenza Season. MMWR Recomm Rep 2021;70(No. RR-5):1-30.

<sup>(h)</sup> These values are based on the clearance of the particular antiviral. LAIV4 should not be administered to persons who have taken oseltamivir or zanamivir within the previous 48 hours, peramivir within the previous 5 days, or baloxavir within the previous 17 days. This “contraindication” is due to concern with reduced effectiveness of the vaccine. To obtain specific information, please refer to Grohskopf LA, Alyanak, E, Broder KR, et. al. Prevention and Control of Seasonal Influenza with Vaccines: Recommendations of the Advisory Committee on Immunization Practices — United States, 2020–21 Influenza Season. MMWR Recomm Rep 2020;69 (No. RR-8:1-26. Also at

<https://www.cdc.gov/mmwr/volumes/69/rr/pdfs/rr6908a1-H.pdf> [PDF](#)

<sup>(i)</sup> This precaution applies to infants younger than 9 months old

<sup>(j)</sup> HIV-infected children may receive varicella vaccine if CD4+ T-lymphocyte count is  $\geq 15\%$  and should receive MMR vaccine if they are aged  $\geq 12$  months and do not have evidence of current severe immunosuppression (i.e., individuals aged  $\leq 5$  years must have CD4+T lymphocyte [CD4] percentages  $\geq 15\%$  for  $\geq 6$  months; and individuals aged  $> 5$  years must have CD4+percentages  $\geq 15\%$  and CD4+ $\geq 200$  lymphocytes/mm<sup>3</sup> for  $\geq 6$  months) or other current evidence of measles, rubella, and mumps immunity. In cases when only CD4+cell counts or only CD4+percentages are available for those older than age 5 years, the assessment of severe immunosuppression can be based on the CD4+values (count or percentage) that are available. In cases when CD4+percentages are not available for those aged  $\leq 5$  years, the assessment of severe immunosuppression can be based on age-specific CD4+counts at the time CD4+counts were measured; i.e., absence of severe immunosuppression is defined as  $\geq 6$  months above age-specific CD4+count criteria: CD4+count  $> 750$  lymphocytes/mm<sup>3</sup> while aged  $\leq 12$  months and CD4+count  $\geq 500$  lymphocytes/mm<sup>3</sup> while aged 1 through 5 years. **Sources:** [\[2\]](#) [\[49\]](#).

<sup>(k)</sup> MMR and varicella-containing vaccines can be administered on the same day. If not administered on the same day, these vaccines should be separated by at least 28 days.

<sup>(l)</sup> If active tuberculosis is suspected, MMR should be delayed. Measles vaccination might suppress tuberculin reactivity temporarily. Measles-containing vaccine can be administered on the same day as tuberculin skin or IGRA testing. If testing cannot be performed until after the day of MMR vaccination, the test should be postponed for  $\geq 4$  weeks after the vaccination. If an urgent need exists to skin test or IGRA, do so with the understanding that reactivity might be reduced by the vaccine.

<sup>(m)</sup> family history of congenital or hereditary immunodeficiency in first-degree relatives (e.g., parents and siblings), unless the immune competence of the potential vaccine recipient has been substantiated clinically or verified by a laboratory

<sup>(n)</sup> For RV1 only, based on latex in product/packaging. Note that anaphylactic allergy to latex is covered in the contraindication and would also be isolated to RV 1 in the case of latex. For more details, see [\[54\]](#).

<sup>(o)</sup> No adverse events associated with the use of aspirin or aspirin-containing products after varicella vaccination have been reported; however, the vaccine manufacturer recommends that vaccine recipients avoid using aspirin or aspirin-containing products for 6 weeks after receiving varicella vaccines because of the association between aspirin use and Reye syndrome after varicella. Vaccination with subsequent close monitoring should be considered for children who have rheumatoid arthritis or other conditions requiring therapeutic aspirin. The risk for serious complications associated with aspirin is likely to be greater in children in whom natural varicella develops than it is in children who receive the vaccine containing attenuated VZV. No association has been documented between Reye syndrome and analgesics or antipyretics that do not contain aspirin.”

TABLE 4-2. Conditions incorrectly perceived as contraindications or precautions to vaccination (i.e., vaccines may be given under these conditions)

| Vaccine                                                                                                                                                                                                                  | Conditions commonly misperceived as contraindications or precautions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General for all vaccines, including DTaP, pediatric DT, adult Td, adolescent-adult Tdap, IPV, MMR, Hib, hepatitis A, hepatitis B, varicella, rotavirus, PCV13, IIV, LAIV, PPSV23, MenACWY, MPSV4, HPV, and herpes zoster | <p>Mild acute illness with or without fever</p> <p>Lack of previous physical examination in well-appearing person</p> <p>Current antimicrobial therapy<sup>(a)</sup></p> <p>Convalescent phase of illness</p> <p>Preterm birth (hepatitis B vaccine is an exception in certain circumstances)<sup>(b)</sup></p> <p>Recent exposure to an infectious disease</p> <p>History of penicillin allergy, other nonvaccine allergies, relatives with allergies, or receiving allergen extract immunotherapy</p> <p>History of GBS<sup>(c)</sup></p>                                                                                                                                                                                                                                                                                                                                    |
| DTaP                                                                                                                                                                                                                     | <p>Fever within 48 hours after vaccination with a previous dose of DTP or DTaP</p> <p>Collapse or shock-like state (i.e., hypotonic hyporesponsive episode) within 48 hours after receiving a previous dose of DTP/DTaP</p> <p>Seizure <math>\leq 3</math> days after receiving a previous dose of DTP/DTaP</p> <p>Persistent, inconsolable crying lasting <math>\geq 3</math> hours within 48 hours after receiving a previous dose of DTP/DTaP</p> <p>Family history of seizures</p> <p>Family history of sudden infant death syndrome</p> <p>Family history of an adverse event after DTP or DTaP administration</p> <p>Stable neurologic conditions (e.g., cerebral palsy, well-controlled seizures, or developmental delay)</p>                                                                                                                                           |
| Hepatitis B                                                                                                                                                                                                              | <p>Pregnancy</p> <p>Autoimmune disease (e.g., systemic lupus erythematosus or rheumatoid arthritis)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| HPV                                                                                                                                                                                                                      | <p>Immunosuppression</p> <p>Previous equivocal or abnormal Papanicolaou test</p> <p>Known HPV infection</p> <p>Breastfeeding</p> <p>History of genital warts</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| IIV                                                                                                                                                                                                                      | <p>Nonsevere (e.g., contact) allergy to latex, thimerosal, or egg</p> <p>Concurrent administration of Coumadin (generic: warfarin) or aminophylline</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| IPV                                                                                                                                                                                                                      | Previous receipt of $\geq 1$ dose of oral polio vaccine                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| LAIV                                                                                                                                                                                                                     | <p>Health-care providers that see patients with chronic diseases or altered immunocompetence (an exception is providers for severely immunocompromised patients requiring care in a protected environment)</p> <p>Breastfeeding</p> <p>Contacts of persons with chronic disease or altered immunocompetence (an exception is contacts of severely immunocompromised patients requiring care in a protected environment)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| MMR <sup>(d),(e)</sup>                                                                                                                                                                                                   | <p>Positive tuberculin skin test</p> <p>Simultaneous tuberculin skin or interferon-gamma release assay (IGRA) testing<sup>(f)</sup></p> <p>Breastfeeding</p> <p>Pregnancy of recipient's mother or other close or household contact</p> <p>Recipient is female of child-bearing age</p> <p>Immunodeficient family member or household contact</p> <p>Asymptomatic or mildly symptomatic HIV infection</p> <p>Allergy to eggs</p>                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| PPSV23                                                                                                                                                                                                                   | History of invasive pneumococcal disease or pneumonia                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Rotavirus                                                                                                                                                                                                                | <p>Prematurity</p> <p>Immunosuppressed household contacts</p> <p>Pregnant household contacts</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Tdap                                                                                                                                                                                                                     | <p>History of fever of <math>\geq 40.5^{\circ}\text{C}</math> (<math>\geq 105^{\circ}\text{F}</math>) for <math>&lt; 48</math> hours after vaccination with a previous dose of DTP or DTaP</p> <p>History of collapse or shock-like state (i.e., hypotonic hyporesponsive episode) within 48 hours after receiving a previous dose of DTP/DTaP</p> <p>History of seizure <math>&lt; 3</math> days after receiving a previous dose of DTP/DTaP</p> <p>History of persistent, inconsolable crying lasting <math>&gt; 3</math> hours within 48 hours after receiving a previous dose of DTP/DTaP</p> <p>History of extensive limb swelling after DTP/DTaP/Td that is not an Arthus-type reaction</p> <p>History of stable neurologic disorder</p> <p>History of brachial neuritis</p> <p>Latex allergy that is not anaphylactic</p> <p>Breastfeeding</p> <p>Immunosuppression</p> |
| Varicella                                                                                                                                                                                                                | <p>Pregnancy of recipient's mother or other close or household contact</p> <p>Immunodeficient family member or household contact<sup>(g)</sup></p> <p>Asymptomatic or mildly symptomatic HIV infection</p> <p>Humoral immunodeficiency (e.g., agammaglobulinemia)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |



| Vaccine | Conditions commonly misperceived as contraindications or precautions                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Zoster  | Therapy with low-dose methotrexate ( $\leq 0.4$ mg/kg/week), azathioprine ( $\leq 3.0$ mg/kg/day), or 6-mercaptopurine ( $\leq 1.5$ mg/kg/day) for treatment of rheumatoid arthritis, psoriasis, polymyositis, sarcoidosis, inflammatory bowel disease, or other conditions<br>Health-care providers of patients with chronic diseases or altered immunocompetence<br>Contacts of patients with chronic diseases or altered immunocompetence<br>Unknown or uncertain history of varicella in a U.S.-born person |

Abbreviations: DT = diphtheria and tetanus toxoids; DTP = diphtheria toxoid, tetanus toxoid, and pertussis; DTaP = diphtheria and tetanus toxoids and acellular pertussis; GBS = Guillain-Barré syndrome; HBsAg = hepatitis B surface antigen; Hib = Haemophilus influenzae type b; HIV = human immunodeficiency virus; HPV = human papillomavirus; IIV = inactivated influenza vaccine; IPV = inactivated poliovirus; LAIV = live, attenuated influenza vaccine; MenACWY = quadrivalent meningococcal conjugate vaccine; MMR = measles, mumps, and rubella; MPSV4 = quadrivalent meningococcal polysaccharide vaccine; PCV = pneumococcal conjugate vaccine; PPSV23= pneumococcal polysaccharide vaccine; Td = tetanus and diphtheria toxoids; Tdap = tetanus toxoid, reduced diphtheria toxoid, and acellular pertussis.

- <sup>(a)</sup> Antibacterial drugs might interfere with Ty21a oral typhoid vaccine, and certain antiviral drugs might interfere with varicella-containing vaccines and LAIV4.
- <sup>(b)</sup> Hepatitis B vaccination should be deferred for infants weighing <2,000 g if the mother is documented to be HBsAg negative. Vaccination should commence at chronological age 1 month or at hospital discharge. For infants born to HBsAg-positive women, hepatitis B immune globulin and hepatitis B vaccine should be administered within 12 hours after birth, regardless of weight.
- <sup>(c)</sup> An exception is Guillain-Barré syndrome within 6 weeks of a dose of influenza vaccine or tetanus-toxoid-containing vaccine, which are precautions for influenza vaccines and tetanus-toxoid containing vaccines, respectively.
- <sup>(d)</sup> MMR and varicella vaccines can be administered on the same day. If not administered on the same day, these vaccines should be separated by at least 28 days.
- <sup>(e)</sup> HIV-infected children should receive immune globulin after exposure to measles. HIV-infected children can receive varicella and measles vaccine if CD4+ T-lymphocyte count is >15% [\[54\]](#).
- <sup>(f)</sup> Measles vaccination might suppress tuberculin reactivity temporarily. Measles-containing vaccine can be administered on the same day as tuberculin skin or IGRA testing. If testing cannot be performed until after the day of MMR vaccination, the test should be postponed for at least 4 weeks after the vaccination. If an urgent need exists to skin test or IGRA, do so with the understanding that reactivity might be reduced by the vaccine.
- <sup>(g)</sup> If a vaccinee experiences a presumed vaccine-related rash 7-25 days after vaccination, the person should avoid direct contact with immunocompromised persons for the duration of the rash.

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